

DBMS LAB-09

ROLL NO: S20230010045

```
mysql> use lab09;
Database changed
mysql> delimiter //
mysql> create function my_fun(value numeric(12,2))
    -> returns int
    -> deterministic
    -> begin
    -> declare r int;
    -> select count(*) into r from account where
    -> balance>value;
    -> return (r);
    -> end//
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> delimiter ;
mysql> select my_fun(500);
+-----+
| my_fun(500) |
+-----+
|          4 |
+-----+
1 row in set (0.07 sec)
```

```
mysql> select balance,my_fun(balance) from account order  
-> by balance;
```

balance	my_fun(balance)
350	6
400	5
500	4
700	2
700	2
750	1
900	0

```
7 rows in set (0.00 sec)
```

EXERCISE:

```
mysql> delimiter //  
mysql> create function level(balance decimal(10,2))  
-> returns varchar(20)  
-> deterministic  
-> BEGIN  
->     DECLARE cust_lvl varchar(20);  
->     If(balance>=900) then  
->         set cust_lvl="PLATINUM";  
->     ELSEIF(BALANCE>=700 AND BALANCE<900)  
->         then set cust_lvl="GOLD";  
->     ELSEIF balance<700 then  
->         set cust_lvl="SILVER";  
->     END IF;  
->     RETURN(cust_lvl);  
-> END//
```

```
Query OK, 0 rows affected, 1 warning (0.01 sec)
```

accountnumber	lvl(balance)
A-305	SILVER
A-102	SILVER
A-101	SILVER
A-215	GOLD
A-222	GOLD
A-217	GOLD
A-201	PLATINUM

7 rows in set (0.01 sec)

```
mysql> delimiter //
mysql>
mysql> create function highest_balance()
-> returns int
-> deterministic
-> begin
->   declare high int;
->   select max(income) into high
->   from employee;
->   return high;
-> end //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> delimiter ;
mysql> select distinct emp_name from employee where income = highest_balance();
+-----+
| emp_name |
+-----+
| Vaish    |
+-----+
1 row in set (0.02 sec)

mysql> select *, experience(doj) from employee;
```

emp_id	emp_name	emp_dept	emp_age	place	income	doj	experience(doj)
2505	peter	Finance	32	Newyork	100000	2002-08-25	22
2506	Mark	HR	32	Callifomia	120000	1980-03-25	44
2507	Donald	Finance	28	Arizona	100000	1995-12-26	29
2508	Obama	Management	35	Florida	500000	1990-10-30	34
2509	Linklon	HR	25	Georgia	25000	2000-08-08	16
2510	Kane	Sales	29	Alaska	30000	2000-01-01	24
2511	Adam	Management	38	Callifomia	54000	2020-10-25	4
2512	Mac	Finance	40	Florida	280000	1970-06-09	54
2513	Manas	Accounts	29	India	600000	1990-12-11	34
2514	Vaish	Accounts	30	India	800000	1989-10-10	35
2515	peter	Finance	32	Newyork	100000	1989-10-10	35
2516	Mark	HR	32	Callifomia	120000	1990-12-11	34
2517	Donald	Finance	28	Arizona	100000	1970-06-09	54
2518	Obama	Management	35	Florida	500000	2020-10-25	4
2519	Linklon	HR	25	Georgia	25000	2000-01-01	24
2520	Kane	Sales	29	Alaska	30000	2000-08-08	16
2521	Adam	Management	38	Callifomia	54000	1990-10-30	34
2522	Mac	Finance	40	Florida	280000	1995-12-26	29
2523	Manas	Accounts	29	India	600000	1980-03-25	44
2524	Vaish	Accounts	30	India	800000	2002-08-25	22

20 rows in set (0.00 sec)

```
mysql> delimiter //
mysql> create function count_more(loan_amt int)
-> returns int
-> deterministic
-> begin
->
->     declare count_val int;
->     select count(*) into count_val
->     from loan
->     where amount >= loan_amt;
->     return count_val;
->
-> end //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> delimiter ;
mysql> select count_more(1000);
+-----+
| count_more(1000) |
+-----+
|                5 |
+-----+
1 row in set (0.03 sec)
```

```

mysql> delimiter //
mysql> create function highest_balance()
    -> returns int
    -> deterministic
    -> begin
    ->
    ->     declare high int;
    ->     select max(income) into high
    ->     from employee;
    ->     return high;
    ->
    -> end //
Query OK, 0 rows affected (0.01 sec)

mysql> delimiter ;
mysql> select distinct emp_name from employee where income = highest_balance();
+-----+
| emp_name |
+-----+
| Vasin    |
+-----+
1 row in set (0.01 sec)

```

ERROR HANDLING:

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE emp_insert(
  -> IN name VARCHAR(20),
  -> IN id INT
  -> )
  -> BEGIN
  -> -- exit if the duplicate key occurs
  -> DECLARE EXIT HANDLER FOR 1062
  -> BEGIN
  -> SELECT CONCAT('Duplicate key (' ,name,',',',id,') occurred') AS
  -> message; END;
  -> -- insert a new row into the EMP table
  -> INSERT INTO emp(emp_id,emp_name)
  -> VALUES(id,name);
  -> -- return the row inserted recently
  -> Select * from emp where emp_id=id;
  -> END $$
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> Call emp_insert('John','10');
```

emp_id	emp_name
10	John

1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

```
mysql> Call emp_insert('mart','20');
```

emp_id	emp_name
20	mart

1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

```
mysql> Call emp_insert ('Jo','30');
```

emp_id	emp_name
30	Jo

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```
mysql> alter table emp  
      -> ADD PRIMARY KEY (emp_name);  
Query OK, 0 rows affected (0.07 sec)  
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> Call emp_insert('mart','20');  
+-----+  
| message |  
+-----+  
| Duplicate key (mart,20) occurred |  
+-----+  
1 row in set (0.00 sec)  
  
Query OK, 0 rows affected (0.00 sec)
```

```

mysql> DELIMITER $$
mysql> CREATE PROCEDURE sampleInsert(IN column1 varchar(30), IN column2 int)
  -> BEGIN
  -> DECLARE EXIT HANDLER FOR 1048
  -> BEGIN
  -> \c
mysql>
mysql> DELIMITER $$
mysql> CREATE PROCEDURE sampleInsert(IN column1 varchar(30), IN column2 int)
  -> BEGIN
  -> DECLARE EXIT HANDLER FOR 1048
  -> BEGIN
  -> SELECT "Trying to populate a non-null column with null value" AS message;
  -> END;
  -> INSERT INTO SAMPLE values(column1, column2);
  -> END$$
Query OK, 0 rows affected (0.05 sec)

mysql> CALL sampleInsert(NULL, 300);
  -> $$
+-----+
| message |
+-----+
| Trying to populate a non-null column with null value |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

```